



AUTOMATED E-COMMERCE PLATFORM

SHOPHUB

Microservices Architecture, CI/CD Pipeline
& Cloud Infrastructure on AWS EKS

A complete DevOps solution for containerized e-commerce deployment

AGENDA

01

System Architecture

Data flow, ingress routing & traffic
management



02

Micro- services

6 specialized services with modern
frameworks



03

Data Layer

PostgreSQL & Redis StatefulSets with
persistent storage



04

CI/CD Pipeline

Jenkins automation: build, test, push
& deploy



05

Cloud Infrastructure

AWS EKS cluster with managed
nodes & storage



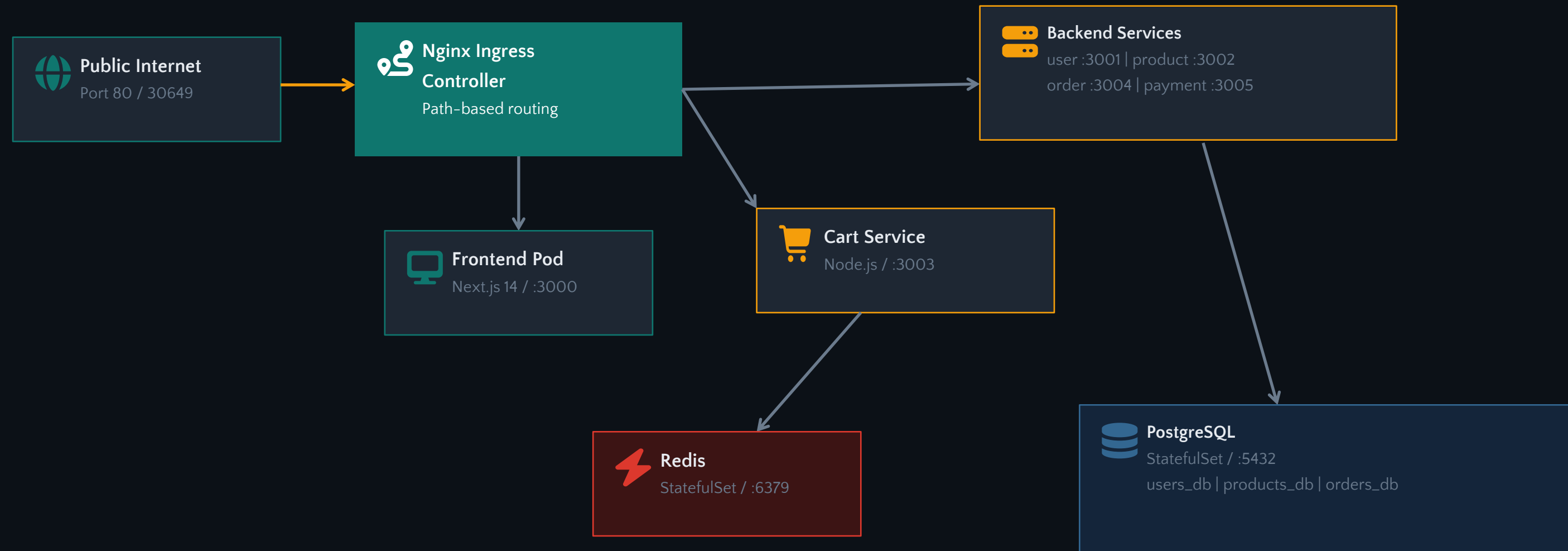


01

SYSTEM ARCHITECTURE

Highly scalable, containerized microservices with intelligent traffic routing

End-to-End Data Flow Architecture



- **Ingress routing** — Path-based traffic distribution
- **Internal services** — Containerized microservice pods
- **Redis cache** — Session/cart data
- **PostgreSQL** — Persistent relational storage

Ingress Routing & Traffic Management



Nginx Ingress Controller

Namespace: ingress-nginx | Ingress Class: nginx

NodePort HTTP: 30649 | Access: 54.90.185.115:30649

Path Prefix	Target Service	Port	Technology
/ (catch-all)	Frontend Pod	3000	Next.js 14 (React + TS)
/api/auth, /api/users	User Service	3001	Node.js + Express + Sequelize
/api/products, /api/categories	Product Service	3002	Python + FastAPI + SQLAlchemy
/api/cart	Cart Service	3003	Node.js + Express + Redis
/api/orders	Order Service	3004	Node.js + Express
/api/payments	Payment Service	3005	Python + FastAPI

⚡ New arrivals every week

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Fast delivery, easy returns.

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Free Shipping over \$50



Secure Checkout



5-Star Reviews

All Products

 Search products...

 All Categories



iPhone 15 Pro

\$999.99

 Add



Samsung Galaxy S24

\$849.99

 Add



MacBook Air M3

\$1299.99

 Add



Sony WH-1000XM5

\$349.99

 Add



Shopping Cart

Clear cart



MacBook Air M3
\$1299.99

\$1299.99

− 1 +



Sony WH-1000XM5
\$349.99

\$349.99

− 1 +



Order Summary

Subtotal (2 items) \$1649.98

Shipping Free

Total \$1649.98

[Proceed to Checkout →](#)

[Continue Shopping](#)

1

Shipping

2

Payment

 Shipping Address

Full Name *

Street Address *

City *

State

ZIP Code *

US

Continue to Payment →

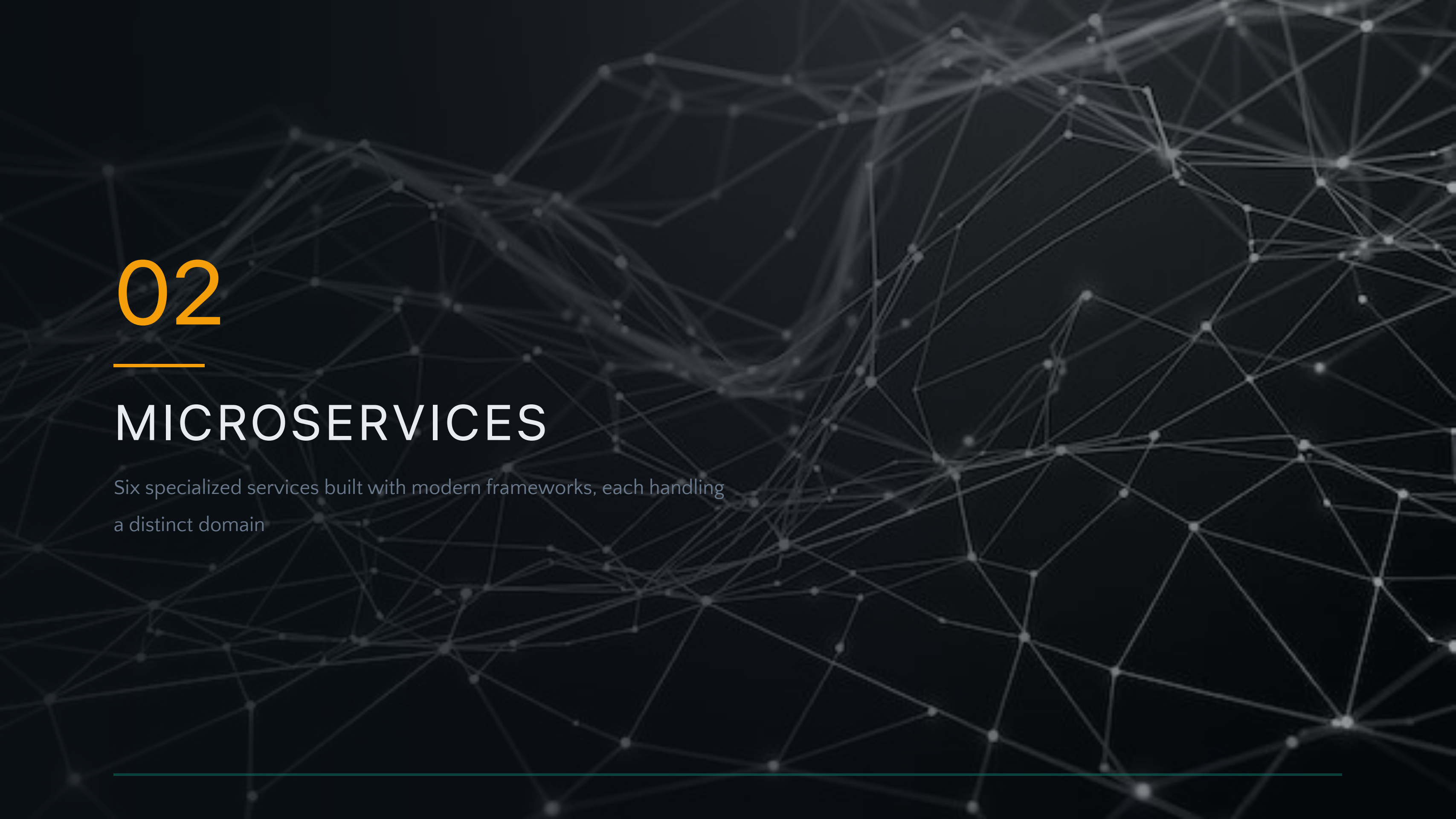
Order Summary

MacBook Air M3 ×1 \$1299.99

Sony WH-1000XM5 ×1 \$349.99

Subtotal \$1649.98

Shipping **Free****Total \$1649.98**



02

MICROSERVICES

Six specialized services built with modern frameworks, each handling a distinct domain

Frontend & User Service

Frontend Service

:3000 Next.js 14 · React · TypeScript

Docker Build

Multi-stage build from `./frontend` context

Routing

Ingress routes `/*` (catch-all) to this service

Purpose

Customer-facing e-commerce UI with SSR/CSR hybrid rendering

User Service

:3001 Node.js · Express · Sequelize

Database

PostgreSQL `users_db`

Core Functions

- User registration & account creation
- Login with credential verification
- JWT token issuance & validation

Routing

Ingress: `/api/auth` & `/api/users`

Security

Stateless JWT auth for all protected routes

Product & Cart Service



Product Service

:3002

Python · FastAPI

Database

PostgreSQL products_table

Core Functions

- Product catalog management
- Category creation & organization
- Item listing with search & filters

Routing

Ingress: [/api/products](#) & [/api/categories](#)

API Style

Modern async Python with auto-generated OpenAPI docs



Cart Service

:3003

Node.js · Express · Redis

Cache Store

Redis StatefulSet :6379

Core Functions

- Active cart item additions
- Quantity updates in real-time
- Cart persistence across sessions

Routing

Ingress: [/api/cart](#)

Performance

Sub-millisecond Redis ops for instant cart updates

Order & Payment Service



Order Service

:3004

Node.js · Express

Database

PostgreSQL `orders_table`

Core Functions

- Shopping cart checkout
- Order creation & validation
- Order history & status tracking

Routing

Ingress: `/api/orders`

Flow

Integrates with Cart & Payment for checkout



Payment Service

:3005

Python · FastAPI

Database

None — stateless transaction processor

Core Functions

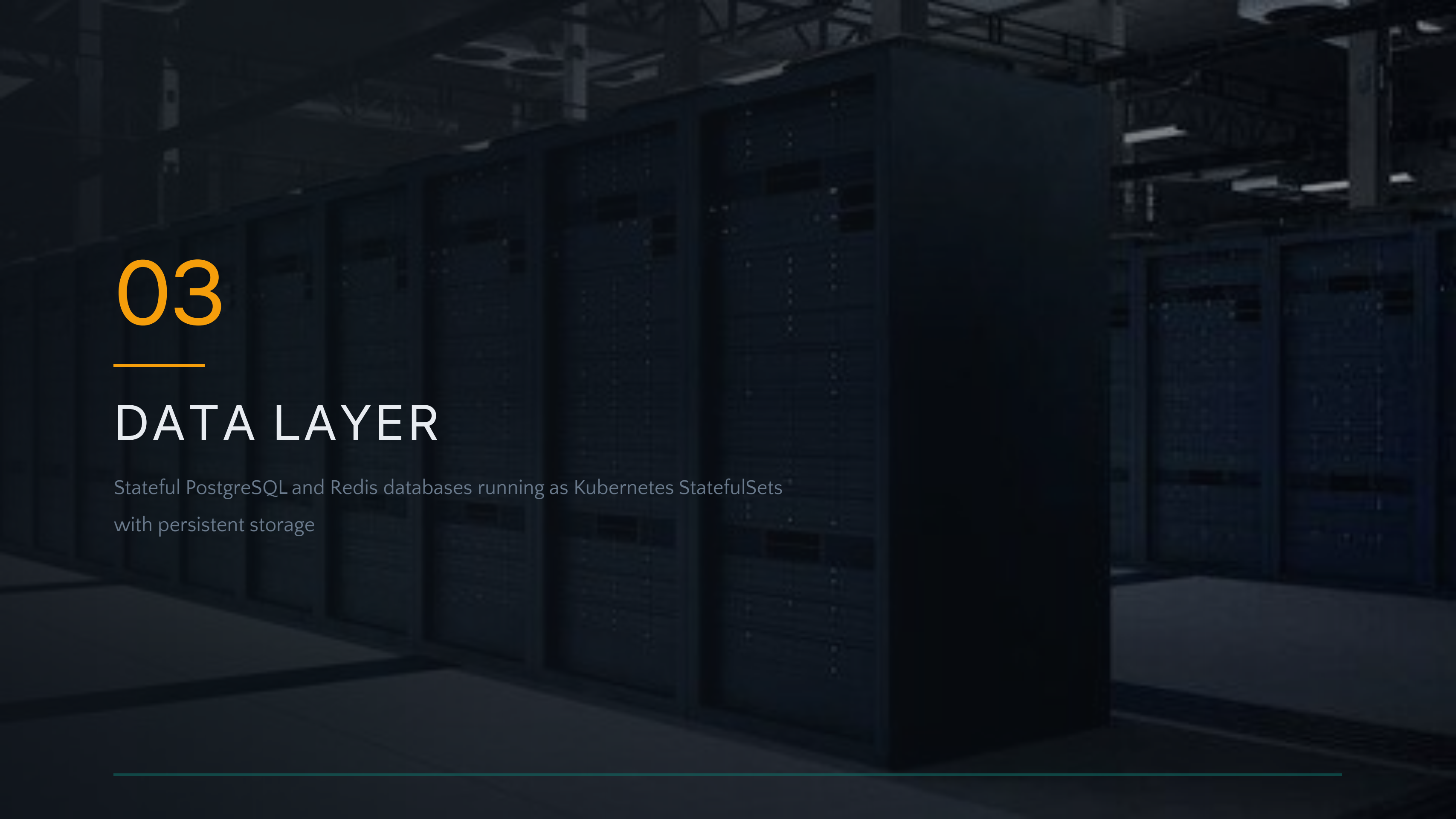
- Mock payment processing
- Transaction validation
- Payment status confirmation

Routing

Ingress: `/api/payments`

Architecture

Lightweight, easily replaceable with real gateway



03

DATA LAYER

Stateful PostgreSQL and Redis databases running as Kubernetes StatefulSets
with persistent storage

PostgreSQL & Redis StatefulSets



PostgreSQL StatefulSet

TCP :5432

Pod `postgres-0` (Single-pod StatefulSet)

Storage

Dynamic 1Gi SSD GP3 volume

Mount: `/var/lib/postgresql/data`

AWS Workaround

PGDATA redirected to avoid EXT4 `/lost+found` boot failures

Tables:

1. users 2. products 3. orders

Service

Headless Service `postgres` for internal discovery



Redis Cache StatefulSet

TCP :6379

Pod `redis-0` (Single-pod StatefulSet)

Storage

Dynamic 1Gi SSD GP3 volume

Mount: `/data`

Purpose

High-performance cache for shopping cart data, sessions, and real-time quantity tracking

Service

ClusterIP Service `redis` for internal access

04

CI/CD PIPELINE

Fully automated Jenkins pipeline triggered by GitHub webhooks for build, test, push, and deploy

Jenkins Pipeline — 4 Stages



Trigger: GitHub webhook push | Credentials: AWS + Docker Hub | Region: us-east-1

01

BUILD



Build Docker images for all 6 services:

- cart-service
- order-service
- payment-service
- product-service
- user-service
- frontend

02

TEST



Spin up all containers and execute the full test suite across services

03

PUSH



Login to Docker Hub using Jenkins credentials

Push all 6 service images with **latest** tag to registry

`docker-hub-credentials` stored in Jenkins

04

DEPLOY



Authenticate kubectl to AWS EKS cluster

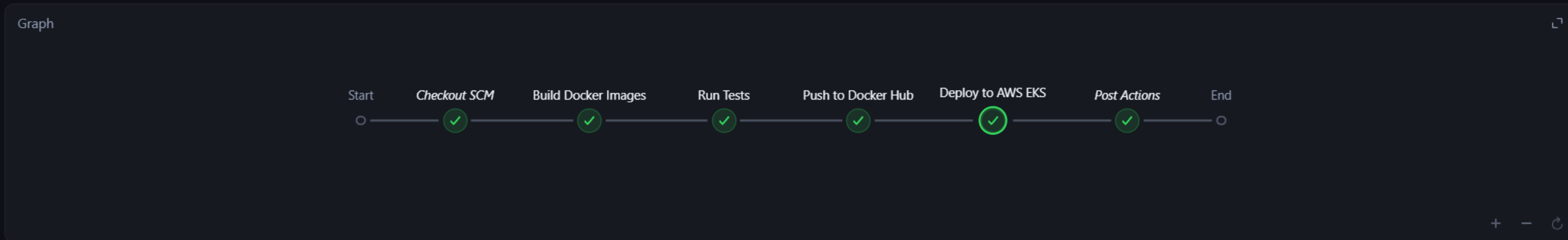
Apply all K8s manifests:

- Deployments
- Services
- StatefulSets
- Ingress rules

✓ < #6

Rerun Settings

Started by GitHub push by AnassEhab33 Started 26 sec ago Queued 8.6 sec Build has been executing for 26 sec Changes



Search

- ✓ Checkout SCM 4s
- ✓ Build Docker Images 13s
- ✓ Run Tests 47s
- ✓ Push to Docker Hub 2m 51s
- ✓ Deploy to AWS EKS 1m 24s
- ✓ Post Actions 1s

✓ Deploy to AWS EKS 1m 24s Started 42m ago Jenkins

Connecting to AWS EKS... 33ms

aws eks update-kubeconfig --name ecommerce-cluster --region us-east-1 9s

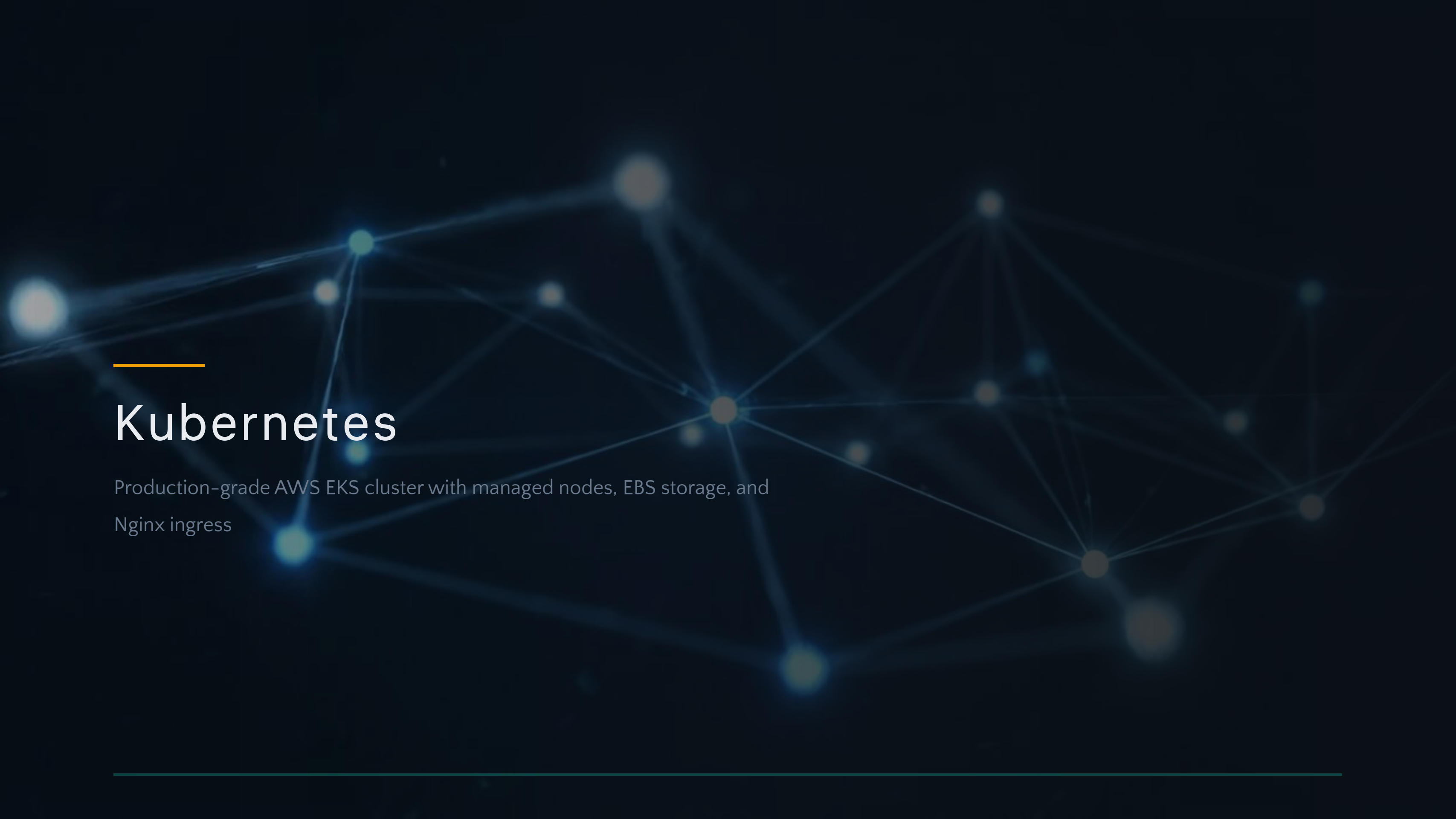
Applying storage class... 2s

kubectl apply -f k8s/gp3-storageclass.yml 23s

```
0 + kubectl apply -f k8s/gp3-storageclass.yml
1 storageclass.storage.k8s.io/gp3 unchanged
```

Applying Kubernetes configs and databases... 9s

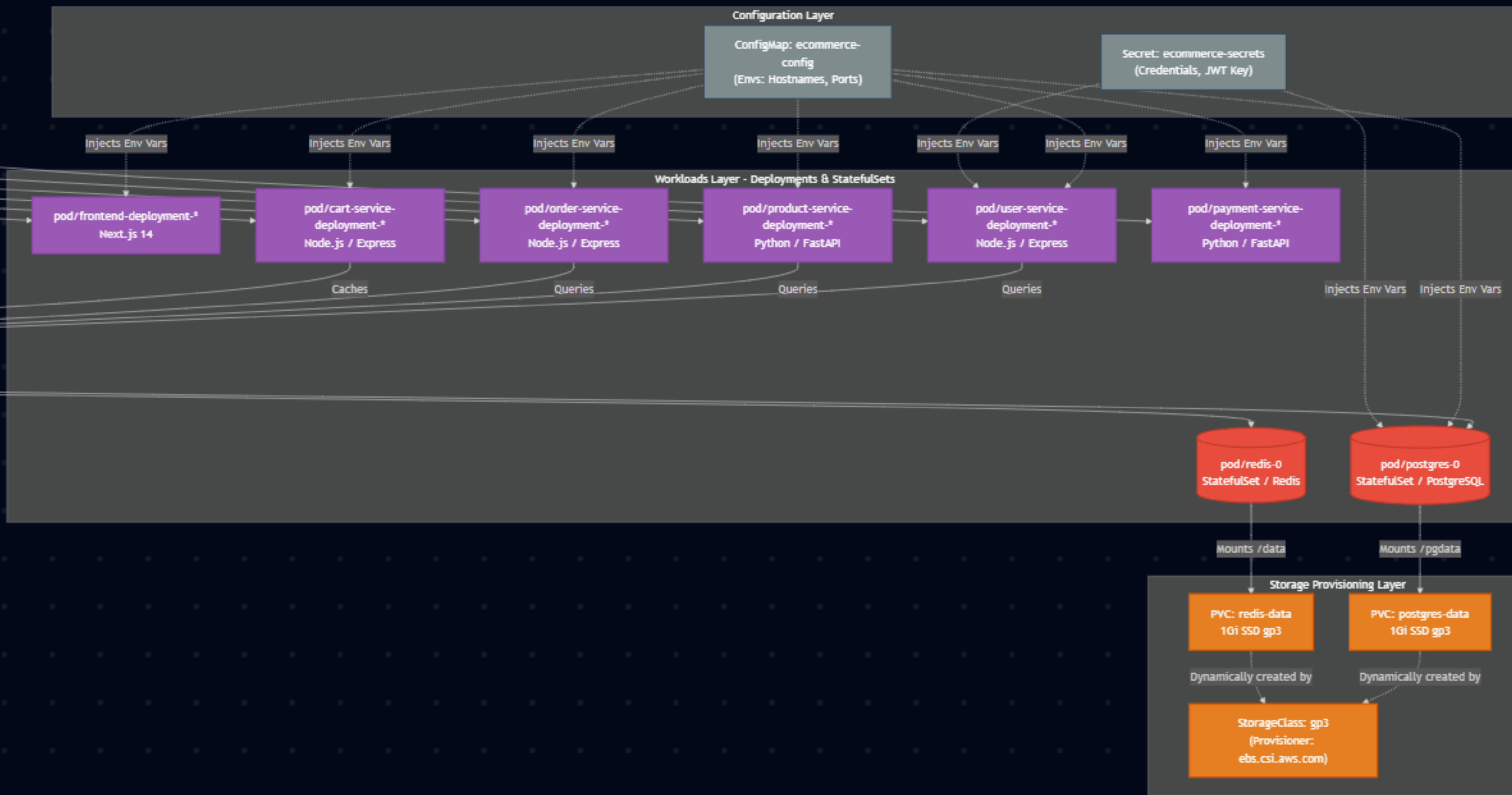
kubectl apply -f k8s/config-map.yml



Kubernetes

Production-grade AWS EKS cluster with managed nodes, EBS storage, and
Nginx ingress





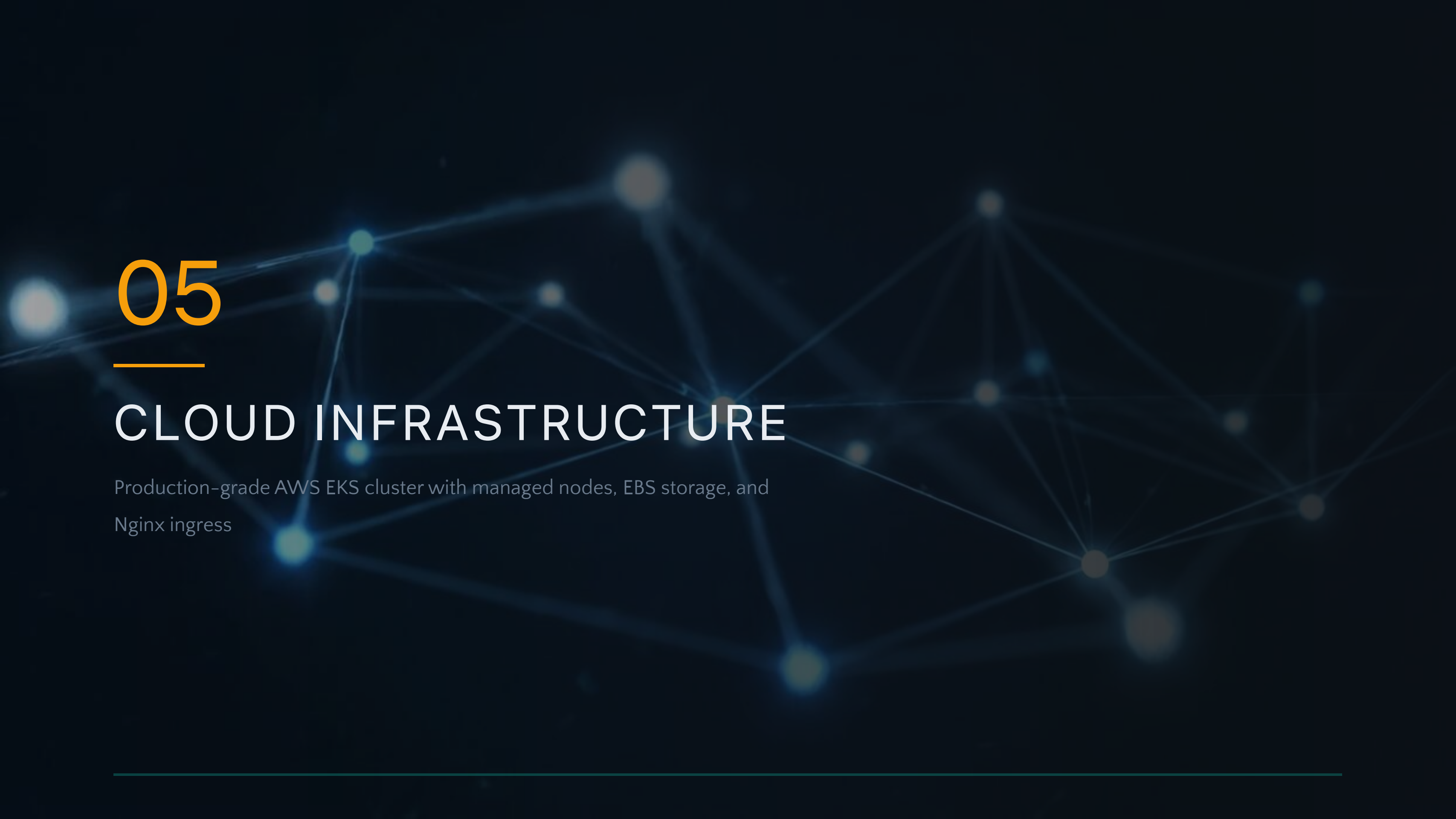
```
PS C:\Users\anass> kubectl get all -n ecommerce
```

NAME	READY	STATUS	RESTARTS	AGE
pod/cart-service-deployment-7bc8b96555-ff9d8	1/1	Running	0	62m
pod/frontend-deployment-76cbdc8f94-shmtv	1/1	Running	0	62m
pod/order-service-deployment-dbc5c665b-lwqdb	1/1	Running	0	62m
pod/payment-service-deployment-64b79cc5ff-lhl9k	1/1	Running	0	62m
pod/postgres-0	1/1	Running	0	3h41m
pod/product-service-deployment-689b5b8855-8dwhh	1/1	Running	0	62m
pod/redis-0	1/1	Running	0	3h43m
pod/user-service-deployment-55f96fb8d9-pnbk8	1/1	Running	0	62m

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/cart-service	ClusterIP	10.100.214.134	<none>	3003/TCP	3h40m
service/frontend	ClusterIP	10.100.163.84	<none>	3000/TCP	3h40m
service/order-service	ClusterIP	10.100.84.51	<none>	3004/TCP	3h40m
service/payment-service	ClusterIP	10.100.135.161	<none>	3005/TCP	3h40m
service/postgres	ClusterIP	None	<none>	5432/TCP	3h43m
service/product-service	ClusterIP	10.100.157.206	<none>	3002/TCP	3h40m
service/redis	ClusterIP	None	<none>	6379/TCP	3h43m
service/user-service	ClusterIP	10.100.244.85	<none>	3001/TCP	3h40m

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
deployment.apps/cart-service-deployment	1/1	1	1	3h41m
deployment.apps/frontend-deployment	1/1	1	1	3h41m
deployment.apps/order-service-deployment	1/1	1	1	3h41m
deployment.apps/payment-service-deployment	1/1	1	1	3h41m
deployment.apps/product-service-deployment	1/1	1	1	3h40m
deployment.apps/user-service-deployment	1/1	1	1	3h40m

NAME	READY	AGE
statefulset.apps/postgres	1/1	3h43m
statefulset.apps/redis	1/1	3h43m



05

CLOUD INFRASTRUCTURE

Production-grade AWS EKS cluster with managed nodes, EBS storage, and Nginx ingress

AWS EKS Cluster Specifications

 EKS Cluster

Cluster Name

ecommerce-cluster

Region

us-east-1 (N. Virginia)

Compute Nodes

2× t3.medium EC2 instances

Managed group: standard-workers

Storage Provider

AWS EBS CSI Driver addon

ebs.csi.aws.com

Storage Class

gp3 (SSD) — default StorageClass

 Ingress & Access

Controller

Ngix Ingress Controller

Namespace: ingress-nginx

Ingress Class

nginx

NodePort Mapping

Public HTTP → Port 30649

Direct Access URL

http://54.90.185.115:30649/

SSL/TLS

Configurable via cert-manager

(not enabled in current setup)

aws

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Service and networking: Services (14)

Service is an abstract way to expose an application running on a set of Pods as a network service. [Learn more](#)

Filter namespaces

Filter Services by name

Advanced query filters

Name	Created
cart-service	4 hours ago
eks-extension-metrics-api	4 hours ago
frontend	4 hours ago
ingress-nginx-controller	4 hours ago
ingress-nginx-controller-admission	4 hours ago
kube-dns	4 hours ago
kubernetes	4 hours ago
metrics-server	4 hours ago
order-service	4 hours ago
payment-service	4 hours ago

Pods

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Workloads: Pods (23)

View details

Pod is the smallest and simplest Kubernetes object. A Pod represents a set of running containers on your cluster. [Learn more](#)

Filter namespaces

Filter Pods by name

► Advanced query filters

<

1

2

3

>

Name	Created	Status
aws-node-h44k2	4 hours ago	Running
aws-node-jstp5	4 hours ago	Running
cart-service-deployment-7bc8b96555-ff9d8	an hour ago	Running
coredns-6976d5bf49-t2qts	4 hours ago	Running
coredns-6976d5bf49-wpr8h	4 hours ago	Running
ebs-csi-controller-9598d6f7d-jfsxv	4 hours ago	Running
ebs-csi-controller-9598d6f7d-l4cbs	4 hours ago	Running
ebs-csi-node-89p9z	4 hours ago	Running

StatefulSets:

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Workloads: StatefulSets (2)

Statefulset manages the deployment and scaling of a set of Pods, and provides guarantees about the ordering and uniqueness of these Pods.

Advanced query filters

Name	Namespace	Type	Created	Pod count	Status
postgres	ecommerce	statefulsets	4 hours ago	1	1 Ready 0 Failed 1 Desired
redis	ecommerce	statefulsets	4 hours ago	1	1 Ready 0 Failed 1 Desired

Config Maps & Secrets

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Config and secrets: ConfigMaps (16)

An API object used to store non-confidential data in key-value pairs. Pods can consume ConfigMap configuration files in a volume.

Learn more

Filter namespaces

Filter ConfigMaps by name

Advanced query filters

Name	Created
amazon-vpc-cni	4 hours ago
aws-auth	4 hours ago
coredns	4 hours ago
ecommerce-config	4 hours ago
extension-apiserver-authentication	4 hours ago
ingress-nginx-controller	4 hours ago
kube-apiserver-legacy-service-account-token-tracking	4 hours ago
kube-proxy	4 hours ago
kube-proxy-config	4 hours ago
kube-root-ca.crt	4 hours ago

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Config and secrets: Secrets (2)

Secret stores sensitive information, such as passwords, OAuth tokens, and ssh keys.

Learn more

Filter namespaces

Filter Secrets by name

Advanced query filters

Name	Created
ecommerce-secrets	4 hours ago
ingress-nginx-admission	4 hours ago

Thank You

A production-ready, automated e-commerce platform demonstrating modern DevOps practices — from microservices to continuous deployment

NEXT.JS · NODE.JS · PYTHON · KUBERNETES · AWS EKS · JENKINS
